[PD27]

오존존데 기반 한반도 대기 오존 특성: 관측/재분석/위성 자료를 활용한 오존 분포

강현규(hyungyu.m.k@gmail.com), 김형규, 오수빈, 구자호[1], 박상서[2], 박주선[3], 이원진[4], 신선아[4], 박진수[5], 김주완

국립공주대학교 대기과학과, 1)연세대학교 대기과학과, 2)울산과학기술원 지구환경도시건설공학과,
3)국립환경과학원 환경위성센터, 4)국립환경과학원 대기과학연구과, 5)부산대학교 환경연구원

1. Introduction

- **Stratospheric ozone** protects life on Earth by absorbing harmful ultraviolet (UV) radiation (*Solomon, 1999*), whereas **tropospheric ozone** acts as a pollutant harmful to human health and vegetation (*Monks et al., 2015*).
- Continuous **monitoring of ozone** distribution is essential to understand its spatial and temporal variability and its impact on air quality and climate (*Schultz et al., 2017*).
- **Ozonesonde observations**, **satellite retrievals**, and **reanalysis datasets** together provide complementary information on the vertical and horizontal structure of atmospheric ozone.
- This study analyzes **ozone variability** over the Korean Peninsula using multi-platform datasets, focusing on transport mechanisms and satellite product evaluation.

3. Data

Table 2. Datasets are used in this study.

Data	Resolution	Reference
ECC Ozonesonde (Electrochemical concentration cell)	100-m level, surface to ~ 7 hPa	Komhyr, 1969
MERRA-2 (Modern-Era Retrospective analysis for Research and Applications, Version 2)	0.5° × 0.625°, 72 hybrid sigma level	Gelaro et al., 2017
EAC4 (ECMWF Atmospheric Composition Reanalysis 4)	0.75° × 0.75°, 60 hybrid sigma level	Inness et al., 2019
ERA5 (ECMWF Reanalysis v5)	1.5° × 1.5°, 37 pressure level	Hersbach et al., 2020
GEMS (Geostationary Environment Monitoring Spectrometer)	7 km × 8 km (gaseous species)	Kim et al., 2020

2. Observation

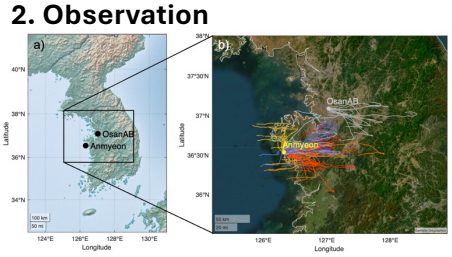


Figure 1. a) Locations of Anmyeon (AMY) and Osan Air Base (OsanAB) sites, where ozonesonde measurements were made during ACCLIP. b) Ascending trajectories of the ozonesondes. The bluish and reddish lines represent the Anmyeon measurements in 2021 and 2022, respectively. Gray lines show OsanAB measurements in 2022.



Figure 2. An example of ozonesonde observation at Yongin station (Lat: 37.33, Lon: 127.26, Alt: 135m) in October 2024.

Table 1. Summary of the ozonesonde measurement sites and records (*Kang et al., 2024*).

Sation	Latitude (°)	Longitude (°)	Elevation (m)	Record (month/year)	Number of profiles
Anmyeon	36.54	126.33	8	Aug / 2021	24
				Jul-Sep / 2022	38
Yongin	37.33	127.26	135	Oct-Nov / 2021	10
				May-Jun / 2022	17
				Aug / 2023	12
				Sep-Oct / 2024	12
				Aug-Sep / 2025	17
OsanAB	37.09	127.03	10	Aug / 2022	11



Figure 3. Electrochemical concentration cell ozonesonde with flight box from EN-SCI corporation.

4. Result 1

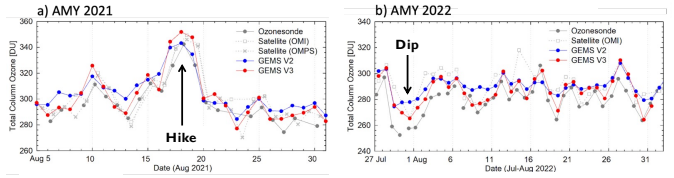
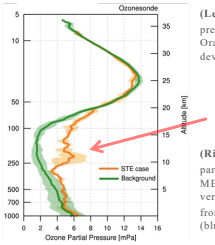
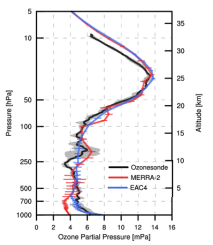


Figure 4. Time series of total column ozone (TCO, unit: DU) at Anmyeon (AMY) for a) 2021 and b) 2022 based on ozonesonde (gray dots with solid lines), satellites (OMI: gray squares, OMPS: gray crosses, GEMS v2: blue dots, and GEMS v3: red dots).



Secondary Ozone Peak
(Park et al., 2012)

(Left) Figure 5. Mean vertical profiles of ozone partial pressure (unit: mPa) from ozonesonde measurements. Orange and green shadings indicate one standard deviation (±1 σ).



(Right) Figure 6. Mean vertical profiles of ozone partial pressure (unit: mPa) from ozonesonde (black), MERRA-2 (red), and EAC4 (blue) for STE case. Mean vertical profiles of ozone partial pressure (unit: mPa) from ozonesonde (black), MERRA-2 (red), and EAC4 (blue)

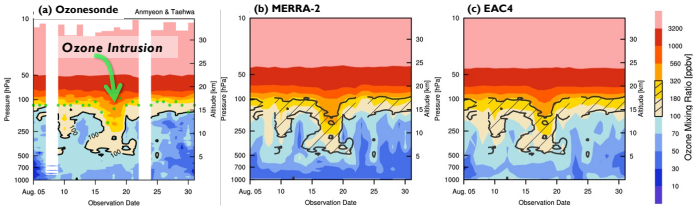


Figure 7. Time-height cross section of ozone mixing ratio (shading, unit: ppbv) from (a) ozonesonde measurements, (b) MERRA-2, and (c) EAC4. The black line indicates 100 ppbv. The green dots represent the tropopause defined by lapse rate. The hatching indicates 100 to 320 ppbv area from ozonesonde measurements.

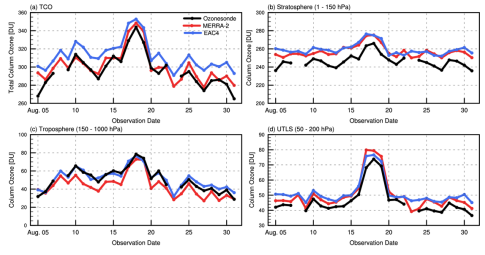


Figure 8. Time series of (a) total column ozone and column ozone in the (b) stratosphere, (c) stratosphere, and (d) upper troposphere-lower stratosphere from ozonesonde measurements (black), MERRA-2 (red), and EAC4 (blue).

- Daily ozonesonde measurements reveal that total column ozone (TCO) variability is closely linked to synoptic-scale weather systems.
- During the summer of 2021, elevated ozone levels were associated with stratosphere–troposphere exchange (STE) events.
- High-resolution vertical ozonesonde profiles enable detailed comparison and validation against reanalysis datasets.

5. Result 2

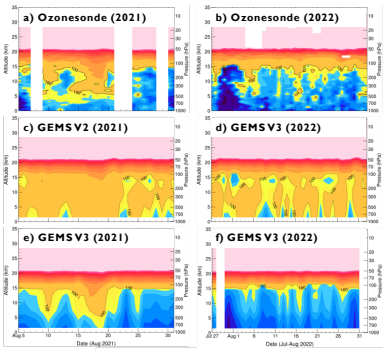


Figure 9. Time-height cross sections of ozone mixing ratio (unit: ppbv) from ozonesonde measurements (a, b), GEMS v2 (c, d), and GEMS v3 (e, f) at Anmyeon site. The left and right panels correspond to measurements from 2021 and 2022, respectively.

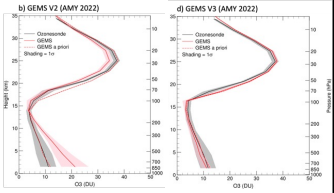


Figure 10. Vertical profiles of the layer-integrated ozone amount (unit: DU) from ozonesondes (black solid lines), GEMS (red solid lines), and GEMS a priori (red dashed lines). The shaded areas represent ±1 standard deviation (σ). The upper and lower panels show results from GEMS v2 and GEMS v3, and the left and right panels correspond to data from 2021 and 2022, respectively. The layer integration is made for ozonesonde data following the layer definition in GEMS data.

- Ozonesonde measurements provide essential reference data for the validation of satellite ozone retrievals.
- Comparison between GEMS O3 Version 2 and Version 3 products shows significant improvements in the troposphere.
- Enhanced ozone distribution patterns were also observed during the 2022 tropical cyclone event, indicating improved satellite performance under dynamic conditions.

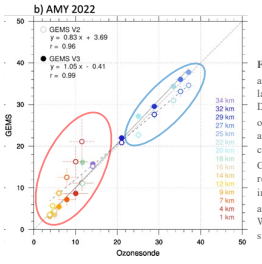


Figure 11. Scatter plots and regression lines of the layer-integrated ozone (unit: DU) from GEMS against ozonesonde for a) 2021 and b) 2022. The open and closed circles represent GEMS v2 and GEMS v3, respectively. Colors indicate the average altitude of the layers. Whiskers denote ±1 standard deviation (σ).

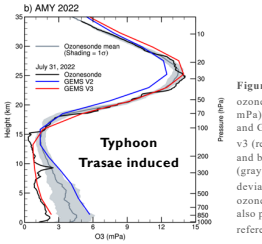


Figure 12. Vertical profiles of ozone partial pressure (unit: mPa) from ozonesonde (black) and GEMS v2 (blue) and GEMS v3 (red) for a) August 18, 2021 and b) July 31, 2022. The mean (gray) and ±1 standard deviation (σ, shading) of the ozonesonde measurements are also presented for each year as a reference.

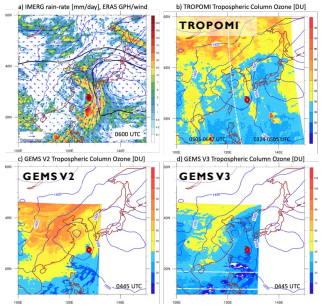


Figure 13. Spatial distributions of a) Integrated multi-satellite retrievals for global precipitation measurement (IMERG) daily rain-rate (shading, unit: mm day^{-1}), 850-hPa GPH (blue contours, unit: m), wind (vectors, unit: m s^{-1}), and 500-hPa GPH (black contours, unit: m) on July 31, 2022.

6. Reference

• Gelaro, R., McCarty, W., Suárez, M. J., Todling, R., Molod, A., Takacs, L., ... & Zhao, B. (2017). The modern-era retrospective analysis for research and applications, version 2 (MERRA-2). Journal of climate, 30(14), 5419-5454. • Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., ... & Thépaut, J. N. (2020). The ERA5 global reanalysis. Quarterly journal of the royal meteorological society, 146(730), 1999-2049. • Inness, A., Ades, M., Agustí-Panareda, A., Barré, J., Benedictow, A., Blechschmidt, A. M., ... & Suttie, M. (2019). The CAMS reanalysis of atmospheric composition. Atmospheric Chemistry and Physics, 19(6), 3515-3556. • Kang, H., Kim, H.-G., Kim, J., Lee, T., Koo, J.-H., Park, S.-S., Choi, Y., Lee, W.-J., Shin, S. A., and Park, J.: Atmospheric and Ozone Profiling Data Measured with Ozonesonde from 2021 to 2024, GEO DATA, 6, 561–577. • Monks, P. S., Archibald, A. T., Colette, A., Cooper, O., Coyle, M., Derwent, R., ... & Williams, M. L. (2015). Tropospheric ozone and its precursors from the urban to the global scale from air quality to short-lived climate forcer. Atmospheric chemistry and physics, 15(15), 8889-8973. • Schultz, M. G., Schröder, S., Lyapina, O., Cooper, O. R., Galbally, I., Petropavlovskikh, I., ... & Zhiqiang, M. (2017). Tropospheric Ozone Assessment Report: Database and metrics data of global surface ozone observations. Elem Sci Anth, 5, 58. • Solomon, S. (1999). Stratospheric ozone depletion: A review of concepts and history. Reviews of geophysics, 37(3), 275-316. • Park, S. S., Kim, J., Cho, H. K., Lee, H., Lee, Y., & Miyagawa, K. (2012). Sudden increase in the total ozone density due to secondary ozone peaks and its effect on total ozone trends over Korea. Atmospheric Environment, 47, 226-235.

Acknowledge: This research was supported by the Ministry of Climate, Energy and Environment (MCEE) of the Republic of Korea (NIER-2022-01-02-045, NIER 2024-01-01-005) and the Specialized university program for confluence analysis of Weather and Climate Data of the Korea Meteorological Institute (KMI) funded by the Korean government (KMA).